

Question				Marking details			Marks available														
							AO1	AO2	AO3	Total	Maths	Prac									
7	(a)			salt A	potassium iodide	accept KI	2		1	3		3									
				observation B	brick red (flame)																
				observations C	apple green (flame) and cream (precipitate)																
				award (1) for each correct row																	
	(b)			<table><tr><td>ion</td><td>Number of protons</td><td>Number of electrons</td></tr><tr><td>Ca²⁺</td><td>20</td><td>18</td></tr><tr><td>Br⁻</td><td>35</td><td>36</td></tr></table>			ion	Number of protons	Number of electrons	Ca ²⁺	20	18	Br ⁻	35	36		2		2		
ion	Number of protons	Number of electrons																			
Ca ²⁺	20	18																			
Br ⁻	35	36																			
				award (1) for each correct row or column no ecf from incorrect number of protons																	
	(c)			Ca ²⁺ + CO ₃ ²⁻ → CaCO ₃ award (1) for reactants award (1) for product (as long as an equation has been attempted) ignore state symbols			2			2											

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			110 (2) if answer incorrect award (1) for either of following 22 530 and 640 ecf possible from subtraction error		2		2	2	2
				Question 7 total	4	4	1	9	2	5